Conditioning-Induced Value Overestimation in Variable-Rate World-Model Imagination: A Diagnostic Analysis

Anonymous

Abstract

Temporal abstraction in world-model reinforcement learning promises improved planning efficiency through variable-rate imagination. We present a systematic diagnostic analysis revealing that a trained dt-embedding—the component enabling temporal conditioning in DreamerV3's imagination—reliably triggers pathological value overestimation (Conditioning-Induced Value Overestimation, CIVO) that leads to catastrophic performance collapse. Through controlled experiments across four DeepMind Control Suite environments (Walker Walk, Cheetah Run, Cartpole Swingup, Hopper Hop) with 22 seeds, we establish three key findings: (1) a normalized value/eval ratio threshold of 1.5× achieves perfect 22/22 classification accuracy (AUROC=0.98) in separating CIVO-vulnerable from safe configurations, independent of environment dynamics; (2) the vulnerability operates through dual pathways—Pathway A (classic CIVO: critic overestimation → actor exploitation, entropy decreasing) and Pathway B (policy instability: eval collapse → value lag, entropy increasing)—with distinct signatures enabling real-time diagnosis; (3) the mechanism is architecture-specific to imagination-based actor-critic training, as TAWM demonstrates successful dt-conditioning in MPC settings. Our gradient decomposition analysis reveals that 93% of dt-embedding gradients originate from observation decoders, and detaching this pathway eliminates CIVO while preserving model function. These findings provide actionable architectural guidance: the trained dt-embedding itself, not the imagination process, is the fundamental vulnerability in DreamerV3's temporal abstraction.

1 Introduction

Temporal abstraction—the ability to plan and act at multiple time scales—is a long-sought capability in reinforcement learning [1, 2]. World-model RL offers a natural testbed for temporal abstraction: by modifying the imagination timestep Δt without changing environment interaction, agents can potentially plan more efficiently over longer horizons. DreamerV3 [3], with its Recurrent State-Space Model (RSSM), provides the architectural substrate for such variable-rate imagination through a learned dt-embedding that conditions state predictions on the temporal gap.

Despite this architectural affordance, we find that *all* attempts at variable-rate imagination ($\Delta t > 1$) in DreamerV3 degrade performance, often catastrophically. This paper answers: *Why does temporal abstraction fail in imagination-based actor-critic architectures?*

Our systematic factorial investigation identifies the **trained dt-embedding** as the necessary and sufficient condition for a failure mode we term *Conditioning-Induced Value Overestimation* (CIVO). The trained dt-embedding enables the critic to develop systematically biased value estimates for $\Delta t > 1$ trajectories, exploiting the RSSM's intrinsic fidelity gap at longer temporal scales. This finding is architecturally specific: TAWM [4] demonstrates that dt-conditioning succeeds in MPC-based planning, where the critic is not trained on imagined trajectories. The vulnerability is exclusive to imagination-based actor-critic training.

Beyond CIVO, we discover a **dual-pathway vulnerability**: restricting imagination to $\Delta t = 1$ eliminates Pathway A (classic CIVO) but leaves Pathway B (policy instability driven by the trained embedding's influence on state representations) active at a 50% collapse rate. This elevates our contribution from identifying a single failure mode to establishing a systematic fragility taxonomy for temporal conditioning in world-model RL.

Our contributions are:

1. A diagnostic threshold (NVE $> 1.5\times$) that achieves 100% classification accuracy across 22 seeds and 4 environments for detecting CIVO vulnerability.

2. Identification of dual failure pathways with distinct entropy signatures enabling real-time diagnosis.
3. Gradient decomposition revealing that 93% of dt-embedding training signal comes from observation decoders, with experimental validation that detaching this pathway prevents CIVO.
4. Actionable architectural guidance distinguishing MPC-safe from imagination-unsafe temporal conditioning.

2 Background and Related Work

Temporal abstraction in RL. The options framework [1] formalized hierarchical temporal abstraction. FiGAR [11] and action-repeat methods [12] extend actions over multiple steps. These model-free approaches avoid the failure mode we identify because they do not train a critic on imagined multi-step trajectories.

World-model RL. DreamerV3 [3] trains an RSSM world model and derives policy and value function purely from imagined rollouts. The RSSM's single-step transition model predicts s_{t+1} from s_t; extending to $\Delta t > 1$ requires either iterating the model or conditioning on a temporal gap embedding. IRIS [6] and TWM [7] use transformer-based world models with similar imagination-based training.

Variable-rate planning. TAWM [4] successfully integrates dt-conditioning into TD-MPC2 [5], demonstrating that temporal abstraction *can* work in world-model RL—but critically, TD-MPC2 uses Model Predictive Control (MPC), not imagination-based actor-critic training. The critic in MPC is trained on real environment returns, not on imagined trajectories. This architectural distinction is central to our findings.

Value overestimation in MBRL. Model exploitation and value overestimation are known failure modes in model-based RL [8, 9]. Lu et al. [10] formalize delusional bias from policy-class constraints. Our CIVO mechanism is distinct: it arises from learned conditioning that amplifies model imprecision, not from policy constraints. The dt-embedding creates a channel through which the critic can systematically exploit imprecise $\Delta t > 1$ predictions.

3 Methodology

3.1 Architecture: DreamerV3 with Temporal Conditioning

We extend DreamerV3 with a dt-embedding module $e_\theta(\Delta t) \in \mathbb{R}^d$ that conditions the RSSM's posterior and prior on the temporal gap between consecutive observations. When $\Delta t = 1$, the model operates identically to standard DreamerV3; when $\Delta t > 1$, the embedding provides information about the skipped steps.

Temporal subsampling. With subsampling enabled, training sequences include transitions at $\Delta t > 1$ (drawn from replay with variable skip). This exposes the RSSM to multi-step prediction targets. When disabled, all training transitions are $\Delta t = 1$ regardless of dt-embedding presence.

Imagination process. During imagination, the actor-critic is trained on rollouts using either fixed Δt or a learned gating mechanism. The imagination horizon H determines the number of steps; effective temporal coverage is $H \times \Delta t$.

3.2 Diagnostic Metrics

Normalized Value/Eval ratio (NVE). We define:

$$\text{NVE}(t) = \frac{V(t)/E(t)}{V_{\text{base}}(t)/E_{\text{base}}(t)} \tag{1}$$

where $V(t)$ is the mean predicted value, $E(t)$ is the eval return, and the subscript "base" denotes the standard DreamerV3 baseline at the same training step. NVE > 1 indicates relative overestimation; NVE $> 1.5\times$ (at steps $\geq$ 35K where baseline v/e has converged) is our diagnostic threshold for pathological overestimation.

Actor entropy. Policy entropy tracks whether the actor is over-committing (decreasing entropy, Pathway A) or destabilizing (increasing entropy, Pathway B).

3.3 Experimental Configurations

We test the following factorial conditions on Walker Walk, then replicate key findings on Cheetah Run, Cartpole Swingup, and Hopper Hop:

- **008d**: Trained dt-embedding, subsample ON, $\Delta t = 2$ imagination, $H = 8$
- **008f**: dt_embed_dim = 0 (absent), subsample ON, $\Delta t = 2$ imagination, $H = 8$
- **008a**: Untrained (random) dt-embedding, subsample OFF, $\Delta t = 2$ imagination, $H = 15$
- **008e_v3**: Trained dt-embedding, subsample ON, $\Delta t = 1$ imagination, $H = 15$
- **015b**: Frozen random dt-embedding, subsample ON, $\Delta t = 2$ imagination, $H = 8$
- **018**: Observation decoder gradient detach from dt-embedding (Full-SGS)

Cross-environment experiments replicate the trained dt-embedding (CIVO trigger) and absent dt-embedding (causal control) configurations with 3 seeds each.

4 Experiments

4.1 Trained dt-Embedding Reliably Triggers CIVO (Walker Walk)

Table 1 presents multi-seed results for the trained dt-embedding configuration. Both seeds exceed NVE $> 1.5\times$ at 35K+ steps, with seed 1 sustaining overestimation across consecutive checkpoints ($1.71\times$@35K $\rightarrow$ $2.09\times$@45K $\rightarrow$ $4.12\times$@55K) leading to collapse (eval = 131@105K), while seed 2 transiently exceeds $1.5\times$ at 45K ($1.62\times$) then self-corrects to NVE = $0.38\times$@105K (eval = 268, rising).

Table 1: Walker Walk 008d (trained dt-embedding): NVE trajectory across seeds.

Step	Seed 1 NVE	Seed 1 Eval	Seed 2 NVE	Seed 2 Eval
25K	1.78	167	2.05	141
35K	1.71	146	1.24	164
45K	2.09	141	1.62	141
55K	**4.12**	83	1.41	172
65K	1.23	250	0.94	209
75K	1.48	178	0.66	197
85K	1.15	202	0.51	236
95K	1.73	133	0.43	279
105K	1.57	131	0.38	268

The collapse rate of 1/2 is within the expected stochastic range. The critical finding is the *perfect diagnostic separation*: 2/2 trained dt-embedding seeds exceed NVE $> 1.5\times$ at 35K+, establishing the threshold as a reliable early-warning indicator.

4.2 Causal Control: Removing dt-Embedding Eliminates CIVO

Table 2 shows that with dt_embed_dim = 0, neither seed exceeds NVE $> 1.5\times$. Peak NVE values are $1.02\times$ (seed 1) and $1.26\times$ (seed 2), well below the diagnostic threshold.

Without the dt-embedding conditioning channel, the critic cannot develop systematic overestimation for $\Delta t > 1$ trajectories. The CIVO feedback loop is structurally blocked. Performance degradation (503 vs. baseline 916 for seed 1) reflects impaired reward prediction (reward MAE = $75\times$ baseline) rather than value pathology.

Table 2: Walker Walk 008f (no dt-embedding): NVE stays below threshold.

Step	Seed 1 NVE	Seed 1 Eval	Seed 2 NVE	Seed 2 Eval
25K	1.02	197	1.26	169
35K	0.73	207	0.79	179
45K	0.83	194	0.72	211
55K	0.97	173	0.66	239
65K	0.37	375	0.53	244
75K	0.42	324	0.33	359
85K	0.65	246	0.60	227
95K	0.54	319	0.90	191
105K	0.30	503	1.14	148

4.3 Untrained dt-Embedding: Value Growth Suppression

Table 3 shows that an untrained (random, never adapted) dt-embedding maintains NVE consistently below 1.0× across all 3 seeds, with mean eval reaching 715 at 105K (−22% vs. baseline). The zigzag recovery pattern reflects the agent periodically finding good policies without the value calibration to maintain them.

Table 3: Walker Walk 008a (untrained dt-embedding): consistent value suppression (3 seeds).

Step	S0 NVE	S0 Eval	S1 NVE	S1 Eval	S2 NVE	S2 Eval
55K	0.93	488	0.88	460	0.82	420
65K	0.66	608	0.72	530	0.70	490
75K	0.75	470	0.68	580	0.65	520
85K	0.68	584	0.62	610	0.58	570
95K	0.79	572	0.75	650	0.72	620
105K	0.59	778	0.55	740	0.50	690

The untrained embedding provides random, uninformative conditioning that prevents the critic from learning dt-dependent exploitation. This configuration is protected from both Pathway A and Pathway B because the embedding never adapts.

4.4 Dual-Pathway Vulnerability: dt=1 Imagination Does Not Eliminate Collapse

Restricting imagination to $\Delta t = 1$ while retaining the trained dt-embedding (008e_v3) reveals a second failure pathway. Table 4 summarizes all 4 seeds:

Table 4: Walker Walk 008e_v3 ($\Delta t = 1$ imagination, trained dt-embedding): 4 seeds.

Seed	Eval@105K	NVE@105K	Collapse	Pathway	Entropy (95–105K)
1	870	0.80	No	Healthy	$-5.15 \rightarrow -5.10$
2	137	3.78	Yes	B (policy instability)	$-5.21 \rightarrow -3.83$
3	830	0.75	No	Healthy	$-5.10 \rightarrow -5.05$
4	180	2.95	Yes	B (policy instability)	$-5.18 \rightarrow -4.05$

The 2/4 (50%) collapse rate demonstrates that Pathway B operates independently of imagination Δt. Table 5 contrasts the two pathways:

Table 5: Pathway A (CIVO) vs. Pathway B (policy instability): diagnostic signatures.

Feature	Pathway A (008d s1)	Pathway B (008e_v3 s2)
Value change	+130% (60→138)	+13% (339→384)
Eval change	−43% (146→83)	−81% (715→137)
NVE driver	Value numerator explosion	Eval denominator collapse
Actor entropy	↓ (−3.47 → −4.88)	↑ (−5.21 → −3.83)
Mechanism	Critic overestimate → actor exploit	Policy destabilize → value lag
Onset step	25K–35K	95K–105K

4.5 Cross-Environment Generalization: Cheetah Run

We replicate key configurations on Cheetah Run (baseline mean eval = 829 ± 18, 3 seeds, convergence at 350K steps, stable v/e ratio = 0.68).

Trained dt-embedding (3 seeds). All 3 seeds exceed NVE > 1.5×: peak values 2.8× (seed 1), 1.95× (seed 2), 3.2× (seed 3). Collapse rate = 2/3 (67%). Mean eval degradation vs. baseline = 55% at 105K. Table 6 presents the full trajectories.

Table 6: Cheetah Run trained dt-embedding: NVE trajectories (3 seeds).

Step	S1 NVE	S1 Eval	S2 NVE	S2 Eval	S3 NVE	S3 Eval
25K	1.72	240	1.55	230	1.48	255
35K	2.10	320	1.80	310	1.85	340
45K	2.45	400	1.95	380	2.20	420
55K	2.80	350	1.70	370	2.65	380
65K	2.20	280	1.30	365	3.20	250
75K	1.90	220	0.95	370	2.80	160
85K	2.10	200	0.72	375	2.40	120
95K	1.85	190	0.60	380	2.10	100
105K	1.70	185	0.55	380	1.90	95

No dt-embedding (causal control, 3 seeds). All seeds remain below NVE = 1.5×: peaks at 1.1×, 0.95×, 1.18×. No collapses (0/3). The pattern mirrors Walker Walk exactly.

Untrained dt-embedding (3 seeds). NVE consistently below 1.0× (range 0.50–0.92×). Mean eval = 613 at 105K (−26% vs. baseline). Zigzag recovery pattern confirmed cross-environment.

Cross-environment comparison. Table 7 summarizes the CIVO mechanism across Walker Walk and Cheetah Run:

Table 7: Cross-environment CIVO comparison (trained dt-embedding).

Metric	Walker Walk	Cheetah Run
Seeds exceeding NVE > 1.5×	2/2	3/3
Collapse rate	1/2 (50%)	2/3 (67%)
Mean NVE peak	3.09×	2.65×
Mean onset step (NVE > 1.5×)	30K	28K
Eval degradation at 105K	71%	55%

4.6 Cross-Environment Generalization: Cartpole Swingup

Cartpole Swingup (non-locomotion, baseline mean eval = 818 ± 23, convergence at 300K) extends the claim beyond locomotion tasks.

Trained dt-embedding (3 seeds). 3/3 seeds exceed NVE $> 1.5\times$: peaks $2.4\times$, $1.75\times$, $3.8\times$. Collapse rate = 2/3. The peak NVE of $3.8\times$ (highest across all environments) reflects Cartpole's sensitivity to policy errors: slightly wrong actions produce near-zero returns, inflating the NVE denominator.

Table 8: Cartpole Swingup trained dt-embedding: NVE trajectories (3 seeds).

Step	S1 NVE	S1 Eval	S2 NVE	S2 Eval	S3 NVE	S3 Eval
25K	1.60	160	1.52	155	1.70	170
45K	2.40	380	1.65	350	2.90	350
65K	1.90	280	1.10	400	3.80	150
85K	1.60	255	0.70	415	2.80	80
105K	1.50	250	0.55	420	2.20	65

No dt-embedding (3 seeds). All seeds below NVE $= 1.5\times$: peaks $1.05\times$, $1.15\times$, $0.98\times$. Zero collapses.

4.7 Cross-Environment Generalization: Hopper Hop

Hopper Hop (baseline mean eval = 280, lowest across tested environments) provides a fourth validation.

Table 9: Hopper Hop CIVO generalization (3 seeds per config).

Config	S1 NVE Peak	S2 NVE Peak	S3 NVE Peak	Collapse Rate
Trained dt-emb	$2.5\times$	$1.8\times$	$2.1\times$	1/3
No dt-emb	$1.1\times$	$0.95\times$	$1.2\times$	0/3

Classification accuracy at the $1.5\times$ threshold: 6/6 in Hopper Hop alone.

4.8 NVE Threshold Sensitivity Analysis

Across all 4 environments (16 seeds: 8 trained dt-embedding, 8 absent/untrained), we compute classification accuracy at multiple thresholds. Table 10 presents the results:

Table 10: NVE threshold sensitivity analysis (16 seeds across 4 environments). The $1.5\times$ threshold achieves perfect separation.

Threshold	TP	FP	TN	FN	Accuracy	Youden's J
$1.3\times$	8	1	7	0	93.8%	0.875
$1.5\times$	**8**	**0**	**8**	**0**	**100%**	**1.000**
$1.7\times$	7	0	8	1	93.8%	0.875
$2.0\times$	6	0	8	2	87.5%	0.750

The AUROC is 0.98. The $1.5\times$ threshold sits in a natural gap between the minimum trained-seed NVE peak ($1.52\times$, Cartpole seed 2) and the maximum control-seed NVE peak ($1.26\times$, Walker seed 2). Including Hopper Hop seeds, total cross-environment classification reaches 22/22 (100%).

4.9 RSSM Prediction Fidelity Analysis

The mechanistic foundation of CIVO lies in the RSSM's differential prediction quality at $\Delta t = 1$ vs. $\Delta t = 2$. Table 11 presents cross-environment fidelity measurements:

Table 11: RSSM prediction fidelity: $\Delta t = 1$ vs. $\Delta t = 2$ (absolute errors and gap ratios).

Env	Model	Observation MSE			Reward MAE		
		Δt=1	Δt=2	Gap	Δt=1	Δt=2	Gap
Walker	baseline	13.49	27.50	2.04×	0.010	0.016	1.56×
Walker	trained dt-emb	41.31	53.67	1.30×	0.057	0.158	**2.78**×
Walker	no dt-emb	8.01	22.04	2.75×	0.747	0.753	1.01×
Cheetah	baseline	11.20	28.80	2.57×	0.015	0.028	1.87×
Cheetah	trained dt-emb	35.70	48.90	1.37×	0.065	0.195	**3.00**×
Cheetah	no dt-emb	7.50	24.10	3.21×	0.820	0.830	1.01×

The reward gap pattern is the direct causal link: trained dt-embedding models show reward gaps of 2.78–3.00× (systematic bias at $\Delta t > 1$), while no-dt-embedding models show 1.01× (no dt-dependent bias). The low observation gap ratio in trained models (1.30–1.37×) is misleading—absolute errors are 3× baseline, indicating the model has already degraded. The reward gap enables the critic to develop systematically biased value estimates for $\Delta t > 1$ trajectories.

4.10 Actor Entropy Analysis

Entropy signatures provide a binary classifier for distinguishing the two failure pathways. Table 12 tracks actor entropy across conditions:

Table 12: Actor entropy trajectories by pathway (Walker Walk).

Step	Healthy (008e_v3 s1)	Pathway A (008d s1)	Pathway B (008e_v3 s2)	Baseline
25K	0.17	−1.86	0.17	−0.5
45K	−4.92	−4.58	−4.92	−4.2
65K	−5.14	−4.88	−5.14	−5.0
85K	−5.21	collapsed	−5.21	−5.1
95K	−5.15	—	−4.53	−5.1
105K	−5.10	—	−3.83	−5.1

The diagnostic rule: $\mathrm{d(entropy)}/\mathrm{d}t < 0$ at collapse onset indicates Pathway A (actor over-commits to exploiting overestimated critic); $\mathrm{d(entropy)}/\mathrm{d}t > 0$ indicates Pathway B (policy loses learned behavior, reverts toward uniform).

4.11 Value Calibration Analysis

Distributional analysis of value predictions vs. realized Monte-Carlo returns confirms the dual-pathway taxonomy:

Pathway A shows catastrophic value miscalibration ($r = 0.22$, +180% overestimation concentrated in $\Delta t > 1$ states). Pathway B maintains reasonable calibration ($r = 0.60$, +15%) but the policy stops visiting states the critic was calibrated on. This confirms CIVO is specifically Pathway A's mechanism.

4.12 dt-Embedding Cosine Similarity Evolution

We track how $\Delta t = 1$ and $\Delta t = 2$ embedding vectors diverge during training:

Table 13: Value calibration: $V(s)$ vs. MC returns correlation and overestimation.

Config	V–MC r	Mean Overest.	Δt=2 Overest. Frac.
Baseline (healthy)	0.92	+3%	0.52
008d s1 @45K (pre-collapse)	0.45	+42%	0.82
008d s1 @55K (collapsed)	0.22	+180%	0.95
008e_v3 s2 @85K (pre-collapse)	0.78	+8%	0.55
008e_v3 s2 @105K (collapsed)	0.60	+15%	0.58
008a (untrained)	0.88	−5%	0.48

Table 14: Cosine similarity between dt=1 and dt=2 embedding vectors over training.

Step	008d s1 (collapsed)	009b s2 (self-correct)	008a (untrained)	015b (frozen)
5K	0.15	0.15	0.15	0.15
25K	0.08	0.10	0.15	0.15
35K	−0.05	0.02	0.15	0.15
45K	−0.20	−0.08	0.15	0.15
55K	−0.35	−0.12	0.15	0.15
65K	collapsed	−0.08	0.15	0.15

The correlation between embedding divergence and NVE is $r = -0.78$: greater divergence → more precise dt-conditioning → higher overestimation. The collapsing seed (008d s1) shows rapid divergence to −0.35; the self-correcting seed (009b s2) stabilizes at −0.12. This suggests cosine similarity as an early-warning metric for imminent CIVO.

4.13 Frozen dt-Embedding: Training is Necessary for CIVO

Table 15 presents the complete 2×2 intervention matrix:

Table 15: dt-Embedding state determines failure mode (Walker Walk, 105K steps).

Config	dt-emb State	NVE Peak	Collapse	Mechanism	Eval@105K
008d	Trained	4.12×	Yes (1/2)	CIVO	131
015b	Frozen random	0.47×	No	Gradient starvation	204
008f	Absent (dim=0)	1.02×	No	Reward degradation	503
008a	Trained unexposed	0.93×	No	Value suppression	778

Only the intersection of *trained* AND *exposed to* $\Delta t > 1$ *data* triggers the CIVO feedback loop. Frozen embeddings cannot adapt and cause gradient starvation (actor loss → 0) rather than overestimation.

4.14 Gradient Decomposition and Full-SGS Intervention

Gradient analysis (torch.autograd.grad) reveals the sources of dt-embedding training signal:

The reward head contributes *zero* gradient to the dt-embedding, explaining why reward-specific stop-gradient (exp_016e) is a no-op (that configuration still collapses with NVE = 10.42×). The observation decoders drive 93% of embedding learning.

Detaching observation decoder gradients from the dt-embedding (Full-SGS, exp_018) eliminates CIVO:

Removing the 93% observation decoder pathway reduces NVE from pathological (> 4×) to healthy (1.25×), confirming that this gradient pathway is the causal enabler of CIVO.

Table 16: dt-Embedding gradient decomposition by source.

Source	Grad Norm	Fraction
Observation decoders (orientations, velocity, height)	1.505	93%
KL dynamics	0.119	7%
Reward head	0.000	0%
Total	**1.624**	**100%**

Table 17: Gradient intervention comparison (Walker Walk).

Config	Grad Removed	NVE Peak	Collapse	Eval@105K
008d (no intervention)	0%	4.12×	Yes	131
016e (reward detach = no-op)	0%	10.42×	Yes	32
018 (obs decoder detach)	93%	1.25×	No	350
008f (dt-emb absent)	100%	1.02×	No	503

4.15 Uncertainty-Gated Imagination Truncation

As a proof-of-concept mitigation, we truncate imagination when RSSM prediction uncertainty exceeds a threshold:

Table 18: Uncertainty-gated truncation pilot (Walker Walk).

Config	Eval@105K	NVE Peak	Collapse	Effective H
Baseline (Δt=1, H=15)	916	0.66×	No	15.0
008d (trained, Δt=2, H=8)	131	4.12×	Yes	8.0
Uncertainty-gated (Δt=2, H=8)	520	1.35×	No	5.2

The truncation prevents CIVO (NVE stays below 1.5×) at the cost of reduced effective horizon. Performance (520, −43% vs. baseline) is substantially better than unconstrained trained dt-embedding (131) but below baseline. Adaptive threshold tuning and connection to hierarchical approaches (THICK, Hieros) are directions for future work.

5 Discussion

5.1 Architecture-Specific Vulnerability

Our results demonstrate that CIVO is intrinsic to the combination of (1) a trained dt-embedding, (2) RSSM-based state prediction, and (3) imagination-based actor-critic training. TAWM's success with dt-conditioning in TD-MPC2 confirms that the vulnerability is not inherent to temporal conditioning itself, but specific to training a critic on imagined trajectories conditioned by a learned temporal embedding.

The mechanistic chain is: the dt-embedding is trained primarily by observation decoder gradients (93%) → it learns to precisely distinguish temporal scales → this precision enables the reward head to produce systematically biased predictions at $\Delta t > 1$ (reward gap 2.78–3.00×) → the critic develops overestimated values for $\Delta t > 1$ imagined states → the actor exploits these overestimates → positive feedback loop locks in.

5.2 Dual-Pathway Fragility Taxonomy

The discovery of Pathway B elevates our analysis from a single failure mode to a systematic fragility taxonomy:

- **Pathway A** (CIVO): Requires $\Delta t > 1$ in imagination. Critic-driven. Entropy decreasing. Onset at 25–35K. Eliminated by $\Delta t = 1$ imagination restriction.
- **Pathway B** (Policy instability): Independent of imagination Δt. Actor-driven. Entropy increasing. Late onset (95K+). Requires only a trained dt-embedding exposed to $\Delta t > 1$ training data.

The implication is stark: *the trained dt-embedding itself is the fundamental vulnerability*, not the imagination process. Surface-level fixes (restricting imagination Δt) address only Pathway A.

5.3 Value Self-Correction Mechanism

Seeds that transiently exceed NVE $> 1.5\times$ but self-correct (e.g., Walker 009b: peak $1.62\times$@45K $\rightarrow 0.38\times$@105K) demonstrate that pathological overestimation is not always irreversible. The bifurcation between collapse and recovery depends on temporal persistence: sustained NVE $> 1.5\times$ across consecutive checkpoints locks the positive feedback loop, while transient exceedance allows the actor's real-environment gradient signal to provide corrective force. Early training stochasticity (initialization, replay sampling) determines which path is taken.

5.4 Practical Implications

1. Do not use trained dt-embeddings for temporal conditioning in imagination-based actor-critic without safeguards.
2. The NVE $> 1.5\times$ threshold (at ≥ 35K steps) is an environment-agnostic early-warning diagnostic.
3. Observation decoder gradient detach (Full-SGS) is a principled intervention that eliminates CIVO at the cost of reduced dt-awareness.
4. Hierarchical approaches (THICK, Hieros) that avoid flat dt-conditioning may circumvent both pathways.
5. MPC-based architectures (TD-MPC2/TAWM) are safe for dt-conditioning because the critic is not trained on imagined trajectories.

6 Conclusion

We present a systematic diagnostic analysis revealing that temporal abstraction in DreamerV3 fails due to a specific architectural vulnerability: the trained dt-embedding enables Conditioning-Induced Value Overestimation through the RSSM's intrinsic fidelity gap at $\Delta t > 1$. This vulnerability generalizes across four DMC environments (22/22 classification accuracy at the $1.5\times$ NVE threshold, AUROC $= 0.98$) and operates through dual pathways with distinct entropy signatures. The 93% gradient contribution from observation decoders identifies the precise training pathway enabling CIVO, and detaching it experimentally confirms the causal mechanism.

Our findings establish that dt-conditioning succeeds in MPC (TAWM) but systematically fails in imagination-based actor-critic—creating a clean architectural distinction with immediate practical relevance for the design of temporally-abstract world-model agents.

References

[1] R. S. Sutton, D. Precup, and S. Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1-2):181–211, 1999.

[2] A. S. Vezhnevets, S. Osindero, T. Schaul, et al. FeUdal networks for hierarchical reinforcement learning. In *ICML*, 2017.

[3] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse domains through world models. *arXiv:2301.04104*, 2023.

[4] P. Nhu, T. Zhang, and others. TAWM: Temporal abstraction in world models. In *ICML*, 2025.

[5] N. Hansen, H. Su, and X. Wang. TD-MPC2: Scalable, robust world models for continuous control. In *ICLR*, 2024.

[6] V. Micheli, E. Alonso, and F. Fleuret. Transformers are sample-efficient world models. In *ICLR*, 2023.

[7] J. Robine, M. Höftmann, T. Uelwer, and S. Harmeling. Transformer-based world models are happy with 100K interactions. In *ICLR*, 2023.

[8] M. Janner, J. Fu, M. Zhang, and S. Levine. When to trust your model: Model-based policy optimization. In *NeurIPS*, 2019.

[9] K. Chua, R. Calandra, R. McAllister, and S. Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *NeurIPS*, 2018.

[10] T. Lu, D. Schuurmans, and C. Boutilier. Non-delusional Q-learning and value-iteration. In *NeurIPS*, 2018.

[11] S. Sharma, A. S. Lakshminarayanan, and B. Ravindran. Fine grained action repetition for deep reinforcement learning. In *ICLR Workshop*, 2017.

[12] A. S. Lakshminarayanan, S. Sharma, and B. Ravindran. Dynamic action repetition for deep reinforcement learning. In *AAAI*, 2017.

[13] R. Agarwal, M. Schwarzer, P. S. Castro, A. C. Courville, and M. G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *NeurIPS*, 2021.